

Complete Technical Dossier

RASTA-AI — AI-Based Smart Logistics and Accessibility
Intelligence Platform for the North Eastern Region

What this document is	The full description of the system: what it is, who it is for, how every part works, what the database holds, what the intelligence layer computes, what is built, and what is honestly not.
Problem Statement	SIH26002 · Ministry of Development of North Eastern Region (MDoNER)
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Companion document: **Complete Technology Stack** — versions, packages, endpoints and configuration. This dossier explains behaviour; that one lists components.

§ Contents

1	What this project is, in plain language	10	GPS, telemetry and truthfulness
2	The problem, in detail	11	Offline and resilience design
3	Users and what each one needs	12	Safety: Fleet Sentinel and escalation
4	System architecture	13	Security, permissions and privacy
5	The database — every table and column	14	The two client applications
6	Enumerations — the complete vocabulary	15	Verification and testing
7	The trip lifecycle, end to end	16	Deployment and environments
8	The intelligence layer, engine by engine	17	Known gaps, stated honestly
9	AI and machine learning	18	Roadmap
		19	Glossary

1 What this project is, in plain language

RASTA-AI is a fleet command platform for freight trucks in India's North East. It does two things a normal GPS tracker does not: it treats roads as *changeably passable* rather than as fixed lines on a map, and it escalates when a driver goes silent.

There are two applications over one backend:

- **A manager web console** — the command picture. Plan a shipment, assign a truck and driver, compare three candidate corridors with their risks and fuel costs, authorise one, dispatch, then watch the fleet and be told about problems.
- **A driver Android app** — the field client and the safety endpoint. See the assigned trip, verify the truck, follow turn-by-turn guidance, log stops, ask questions in Assamese, Bengali, Hindi or English, and be found if something goes wrong.

Everything a client reads or writes goes through one FastAPI backend. No client holds a database credential. That is not a stylistic preference — it is the reason the capacity rules, document enforcement and the safety escalation cannot be bypassed by a modified client.

THE ONE-SENTENCE PRODUCT VISION

The manager should never have to ask "where is that truck and is the driver okay?" — the system should have already answered both, and if it cannot, it should have already escalated.

The three capabilities that distinguish this from a GPS tracker

Capability	What it means concretely
Accessibility intelligence	The road network is modelled as changeably passable. Weather readings, landslide exposure and closure evidence continuously revise which corridors are usable, and an active trip is re-assessed against that revision. A route that cannot be assessed is reported as <i>unassessed</i> , not as safe.
Predictive fuel	Consumption is estimated from payload, cumulative elevation climb and cruise speed using a published physics formula, so a 300 km route through the Barail hills and a 300 km route along the Brahmaputra valley are not treated as equivalent. CO ₂ is reported alongside litres.
Fleet Sentinel	A deterministic safety net that escalates on driver <i>silence</i> — which is exactly the failure mode that terrain and poor coverage produce. It contains no language model anywhere in its decision path, by written policy.

What it is explicitly not

- Not a marketplace, load board or freight broker.
- Not a replacement for emergency services — it escalates to a human manager, who decides.
- Not a payments system. Financial state is *tracked*; no money moves.
- Not an autonomous router. A truck's route changes only when a person accepts the change.

2 The problem, in detail

Freight movement in the North East is structurally harder than in the rest of India, and the difficulty is not primarily a vehicle problem — it is an **information** problem.

2.1 Physical constraints

Constraint	Consequence for logistics
The Siliguri Corridor	All eight NER states are reached through a land bridge roughly 22 km wide — the "Chicken's Neck". A single blockage there affects eight states at once.
Mountainous terrain	Most inter-state freight moves on two-lane national highways cut into hillsides, where there is frequently no parallel alternative road within 100+ km . This is why the system must be able to say "there is nothing better" rather than always proposing an alternative.
The monsoon	Roughly May–September, landslides, washouts and flooding close highways with little warning and for unpredictable durations. Reopening is not announced in any machine-readable form.
Intermittent coverage	Mobile network coverage along these corridors is patchy. A truck can be genuinely unreachable for hours in entirely normal operation — so "no signal" cannot be treated as an anomaly.
Gradient	Climb, not distance, dominates loaded-truck fuel consumption on hill routes. It is not modelled by any conventional dispatch tool.

2.2 Operational consequences today

- A fleet manager typically learns a truck is stuck when the **driver phones in** — which requires signal, and requires the driver to be able to make the call.
- Route decisions are made from memory and word-of-mouth road reports, not from data.
- Fuel is budgeted per trip by rule of thumb. Gradient and load are not modelled at all.
- When a driver stops responding there is no systematic escalation. Detection depends on somebody noticing an absence.
- Nothing retains a record of which corridors were passable on which date, so the same knowledge is re-learned every monsoon.

THE GAP THIS PRODUCT CLOSES

The manager's picture of the fleet is delayed, partial and verbal. Everything downstream — rerouting, fuel planning, driver safety — inherits that delay.

3 Users and what each one needs

User	Context	What they need
Fleet manager	Desk, office, browser, multiple screens	One live picture of every truck; to be <i>told</i> about problems rather than having to ask; enough evidence behind a recommendation to defend the decision afterwards

User	Context	What they need
Driver	Cab of a truck, phone only, one hand, poor signal, sometimes at night	Very few, very large controls; guidance that keeps working with no network; to be found if something goes wrong; their own language
Authorised reviewer	Senior operations role	The authority to release a route that the automatic gate has held, with the reason recorded against their name
Operations / finance	Back office	Trip cost accuracy, payment and document status
MDoNER / state transport	Policy and infrastructure	An auditable record of which NER corridors were actually passable, and when

The manager and driver applications have genuinely different design constraints, which is why they are two applications over one backend rather than one responsive web app.

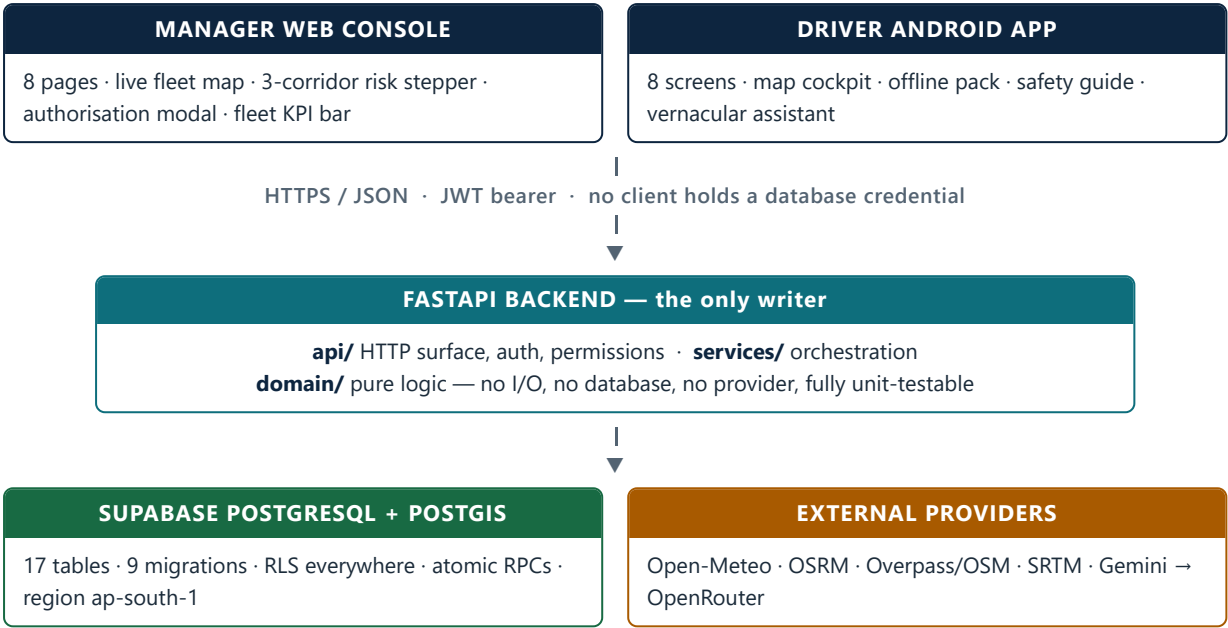
3.1 Manager workflow

1. Opens the dashboard and sees the fleet: active / idle / delayed / stuck, plus any open SOS.
2. Creates a shipment — cargo, weight, pickup, destination, priority.
3. Assigns a truck. **Capacity is validated deterministically** and over-capacity is refused, not warned about.
4. Assigns a driver. If it is a truck the driver has not driven before, a verification step is required.
5. Reviews the proposed corridors: primary, fuel-efficient and emergency backup, each with distance, ETA, estimated fuel and identified risk — and each with the evidence gaps named.
6. Authorises and selects one. Where the automatic gate holds a route, an authorised reviewer must issue a single-use token with a written rationale.
7. Dispatches. From here the manager is a monitor, not an operator.
8. Receives alerts, not questions: road deteriorates → affected trips identified → reroute proposed or an alert raised if there is nothing better.
9. If a truck goes stationary and the driver does not respond, receives a fully populated SOS briefing with everything needed to act.

3.2 Driver workflow

1. Logs in; sees the assigned trip.
2. On a new truck: photographs it and confirms registration, odometer, fuel level and visible damage.
3. Accepts and starts the trip. The phone streams GPS, buffering when offline.
4. Follows the corridor with turn-by-turn guidance and spoken instructions.
5. Receives route changes and hazard warnings as they are issued — never silently.
6. If stationary too long, answers one question with one large button: **Are you safe?**
7. Logs stops and captures proof of delivery at the destination.

4 System architecture



4.1 The layering rule

Layer	May depend on	Responsibility
api/	services, schemas, core	HTTP shape, authentication, permission checks, error mapping. No business rules.
services/	models, domain, providers	Orchestration: load from the database, call providers, invoke domain logic, persist and audit.
domain/	nothing	Pure functions over dataclasses. No SQLAlchemy, no httpx, no FastAPI import anywhere in this package.
models/	SQLAlchemy	Tables, columns, constraints, enums.
core/	—	Configuration, security primitives, permission constants, rate limiting, error types.

WHY THIS LAYERING MATTERS HERE SPECIFICALLY

Because `domain/` touches nothing external, the risk weights, eligibility policy, fuel coefficients and road-memory state machine can be exercised exhaustively in milliseconds with no database and no network. That is what makes it reasonable to publish the constants — they are individually testable, so they can be argued with and tuned rather than defended as a black box.

4.2 Two performance rules the services layer follows

- **The database connection is released before an external provider is called.** A request that holds a pooled connection across a network call spends the database budget waiting on somebody else's server. Route risk fans out to several weather requests, so the window is long.
- **`commit`, not `rollback`, is used to release it.** Both end the transaction, but `rollback` expires every object in the session — including the actor the permission dependency loaded — and the next attribute access then raises `MissingGreenlet`.

4.3 Atomicity

Operations that must be one consistent state change are implemented as single server-side calls, not as several client-side writes:

Operation	Why it must be atomic
plan_trip	Creating a shipment, then a trip, then stops cannot be atomic across a network. A failure between the two leaves an orphaned shipment. One RPC, one transaction.
dispatch_trip	Status change plus event plus notification must not partially apply.
select_route	The chosen route must become SELECTED, the previous one must be demoted, and the trip pointer must move — together. A failure between those writes would leave a trip pointing at a route whose state disagrees, which is the exact inconsistency the atomic-switch requirement exists to prevent.

5 The database — every table and column

PostgreSQL 17.6 with PostGIS 3.3, hosted by Supabase in `ap-south-1` (Mumbai). 17 tables across 9 Alembic migrations. Row Level Security is enabled on every table with no policies, because Supabase publishes the `public` schema through its Data API — a table without RLS would be readable by anyone holding the anon key.

5.1 Identity and access

Table	Columns
users	id · email · phone · password_hash · role · display_name · is_active · last_login_at
refresh_tokens	id · user_id · token_hash · family_id · issued_at · expires_at · revoked_at · revoked_reason · replaced_by_id · user_agent · ip_address
audit_logs	id · actor_user_id · action · entity_type · entity_id · before · after · reason · ip_address · created_at

Note on `refresh_tokens`: only a hash is stored, never the token. `family_id` plus `replaced_by_id` form the rotation chain that makes reuse detection possible — presenting an already-replaced token revokes the entire family.

5.2 People

Table	Columns
drivers	id · user_id · full_name · photo_url · phone · emergency_contact_name · emergency_contact_phone · licence_number · licence_class · licence_expiry · date_of_joining · status · base_salary_monthly
driver_documents	id · driver_id · doc_type · doc_number · file_url · issued_on · expires_on · status · verified_by · verified_at · created_at

`base_salary_monthly` is gated behind the separate `driver:read_sensitive` permission — a manager who can dispatch does not automatically see payroll.

5.3 Fleet

Table	Columns
trucks	id · registration_number · photo_url · truck_type · make · model · manufacture_year · max_capacity_kg · current_load_kg · axle_count · height_m · length_m · fuel_tank_capacity_l · baseline_mileage_kmpl · odometer_km · status
truck_documents	id · truck_id · doc_type · doc_number · file_url · issued_on · expires_on · status · verified_by · verified_at · created_at
truck_maintenance	id · truck_id · event_kind · description · odometer_km · cost · performed_on · next_due_on · created_at

Table	Columns
driver_truck_assignments	id · driver_id · truck_id · assigned_by · status · assigned_at · verification_photo_url · reported_registration · reported_odometer_km · reported_fuel_level_pct · reported_damage_notes · verified_at · mismatch_flagged · manager_review_note · ended_at

`max_capacity_kg`, `height_m`, `length_m` and `axle_count` are what make deterministic capacity refusal and (in future) truck-restriction routing possible. `baseline_mileage_kmpl` is the permanent fallback for the fuel model. A partial unique index enforces **one active assignment per driver**.

5.4 Cargo and shipments

Table	Columns
shipments	id · reference_code · client_name · client_contact · pickup_address · pickup_location · destination_address · destination_location · total_weight_kg · priority · status · scheduled_pickup_at · expected_delivery_at · created_by
cargo_items	id · shipment_id · cargo_type · cargo_name · weight_kg · quantity · is_hazardous · is_perishable · handling_notes

The two **bold** columns are PostGIS `geography(Point,4326)`. An address with no resolved coordinate cannot be planned over — the manager UI refuses an unlocated endpoint rather than silently planning from a string.

5.5 Trips and execution

Table	Columns
trips	id · trip_code · shipment_id · truck_id · driver_id · assignment_id · status · selected_route_id · dispatched_at · driver_accepted_at · driver_accepted_by · started_at · delivered_at · closed_at · planned_eta · current_eta · delay_minutes · created_by
trip_stops	id · trip_id · sequence · kind · status · name · address · location · geofence_radius_m · planned_arrival_at · actual_arrival_at · actual_departure_at · notes
trip_routes	id · trip_id · kind · state · geometry · distance_km · estimated_duration_min · estimated_fuel_litres · estimated_fuel_cost · fuel_estimate_source · risk_score · risk_factors · maneuvers · routing_provider · provider_route_id · superseded_by · created_at
trip_events	id · trip_id · kind · description · payload · location · actor_user_id · occurred_at
gps_points	id · trip_id · driver_id · truck_id · location · altitude_m · speed_kmph · heading_deg · accuracy_m · device_fix_id · recorded_at · received_at · is_mock_location

DETAILS THAT MATTER

- `trip_routes.geometry` is a PostGIS `geography(LineString,4326)` — the actual corridor, not a straight line between endpoints.
- `fuel_estimate_source` records *how* the number was produced (CALIBRATED, ESTIMATED, BASELINE_ONLY, UNAVAILABLE). A litre figure without its provenance is not usable evidence.
- `risk_factors` stores the component breakdown and the named unavailable factors, so the score is reconstructible after the fact.
- `superseded_by` preserves the chain when a corridor is replaced — history is never overwritten.
- `gps_points.device_fix_id` makes ingestion **idempotent on (trip_id, device_fix_id)**. A driver whose phone replays a buffered queue after a dead zone cannot create duplicate history.
- `gps_points.recorded_at` vs `received_at` are kept separately, which is the only way to distinguish "the truck was there then" from "we heard about it then".
- `is_mock_location` is stored rather than filtered, so a spoofed fix is visible in the record instead of quietly discarded.

- `trip_events` is append-only and is the audit trail of the trip itself.

5.6 Route review authorisation

Table	Columns
<code>route_review_authorizations</code>	<code>id</code> · <code>trip_id</code> · <code>route_id</code> · <code>evidence_digest</code> · <code>evidence_snapshot</code> · <code>policy_version</code> · <code>evidence_version</code> · <code>basis</code> · <code>route_state_at_issue</code> · <code>reviewer_user_id</code> · <code>rationale</code> · <code>issued_at</code> · <code>expires_at</code> · <code>consumed_at</code> · <code>consumed_by_user_id</code> · <code>revoked_at</code> · <code>revoked_by_user_id</code> · <code>revoked_reason</code>

This is the most carefully designed table in the schema, and it is worth explaining why every column is there. When the automatic gate holds a route — because landslide evidence is unknown, or because a high hazard was reported — a human with the `route:review_authorize` permission may release it. That release is:

- **Single-use** — `consumed_at` and `consumed_by_user_id` are set on selection, so one authorisation cannot release two routes.
- **Time-boxed** — 30 minutes via `expires_at`. Evidence goes stale.
- **Bound to the evidence it was issued against** — `evidence_digest` plus `evidence_snapshot`. If the weather or hazard picture changes after the token is issued, the token no longer matches what the reviewer actually saw.
- **Versioned** — `policy_version` and `evidence_version`, so an authorisation issued under an older policy is identifiable later.
- **Attributed and reasoned** — `reviewer_user_id` and a written `rationale`. After an incident, "who released this route and on what basis" has an answer.
- **Revocable** — with actor and reason.

5.7 Spatial operations in use

Operation	Where it is used
<code>geography(Point,4326)</code>	Shipment endpoints, stops, GPS fixes, trip events
<code>geography(LineString,4326)</code>	Route corridors
<code>ST_DWithin</code>	Geofence arrival detection at stops; stationary-vehicle detection for Fleet Sentinel
Corridor sampling	Weather is requested at points along the route polyline, not at the endpoints — a corridor is not a point
Partial index <code>ix_trips_active</code>	Matches <code>IN_TRANSIT_STATES</code> exactly, so the Sentinel query scans only trucks actually on the road

6 Enumerations — the complete vocabulary

Every state in the system is a named enum, not a free string. This is what makes the reason-code internationalisation approach possible: the backend sends codes, and the client renders the sentence in the driver's language — no translation model required.

Enum	Values
<code>UserRole</code>	<code>ADMIN</code> · <code>MANAGER</code> · <code>DRIVER</code> · <code>AUTHORISED_REVIEWER</code>
<code>DriverStatus</code>	<code>AVAILABLE</code> · <code>ON_TRIP</code> · <code>OFF_DUTY</code> · <code>SUSPENDED</code>
<code>DocumentStatus</code>	<code>VALID</code> · <code>EXPIRING_SOON</code> · <code>EXPIRED</code> · <code>MISSING</code> · <code>REJECTED</code>
<code>DriverDocumentType</code>	<code>DRIVING_LICENCE</code> · <code>AADHAAR</code> · <code>PAN</code> · <code>POLICE_VERIFICATION</code> · <code>MEDICAL_CERTIFICATE</code> · <code>OTHER</code>
<code>TruckStatus</code>	<code>AVAILABLE</code> · <code>ON_TRIP</code> · <code>MAINTENANCE</code> · <code>BREAKDOWN</code> · <code>RETIRED</code>

Enum	Values
TruckDocumentType	REGISTRATION_CERTIFICATE · INSURANCE · FITNESS_CERTIFICATE · POLLUTION_CERTIFICATE · NATIONAL_PERMIT · STATE_PERMIT · OTHER
MaintenanceKind	SERVICE · REPAIR · BREAKDOWN · INSPECTION
AssignmentStatus	PENDING_VERIFICATION · ACTIVE · ENDED · REJECTED
CargoPriority	LOW · NORMAL · HIGH · CRITICAL
ShipmentStatus	DRAFT · PLANNED · IN_TRANSIT · DELIVERED · CANCELLED
TripStatus	DRAFT · ASSIGNED · VERIFICATION_PENDING · MANAGER_REVIEW · ACTIVE · DELAYED · INCIDENT · DELIVERED · CLOSED · CANCELLED
TripStopKind	PICKUP · DROPOFF · REST · FUEL · CHECKPOINT · OTHER
TripStopStatus	PENDING · ARRIVED · COMPLETED · SKIPPED
RouteKind	PRIMARY · FUEL_EFFICIENT · EMERGENCY_BACKUP
RouteState	PROPOSED · SELECTED · SUPERSEDED · REJECTED_BLOCKED
TripEventKind	CREATED · ASSIGNED · VERIFIED · DISPATCHED · ACCEPTED · STARTED · STOP_ARRIVED · STOP_COMPLETED · ROUTE_CHANGED · DELAY_DETECTED · COMMS_LOST · COMMS_RESTORED · BREAKDOWN_REPORTED · INCIDENT_OPENED · INCIDENT_RESOLVED · DELIVERED · CLOSED · CANCELLED
AuditAction	CREATE · UPDATE · DELETE · STATUS_CHANGE · LOGIN · LOGIN_FAILED · DOCUMENT_ACCESS
RouteReviewBasis	HAZARD_DATA_UNKNOWN · HIGH_HAZARD_REPORTED

TWO ENUM VALUES WORTH POINTING AT

TripEventKind.COMMS_LOST and COMMS_RESTORED exist because losing contact is a *fact about the trip* that belongs in its history, not a UI state. A trip that went dark for two hours in a hill pass should still say so a week later.

RouteState.REJECTED_BLOCKED is distinct from a merely low-scoring route. A blocked corridor is not a bad option — it is not an option.

7 The trip lifecycle, end to end

7.1 The state machine

Trip status transitions are enforced by `domain/trip_state.py`, not by convention. Any transition not listed is refused with the legal targets named in the error.

From	May move to
DRAFT	ASSIGNED · CANCELLED
ASSIGNED	VERIFICATION_PENDING · ACTIVE · CANCELLED
VERIFICATION_PENDING	ASSIGNED · MANAGER_REVIEW · CANCELLED
MANAGER_REVIEW	ASSIGNED · CANCELLED
ACTIVE	DELAYED · INCIDENT · DELIVERED · CANCELLED
DELAYED	ACTIVE · INCIDENT · DELIVERED · CANCELLED
INCIDENT	ACTIVE · DELAYED · CANCELLED
DELIVERED	CLOSED
CLOSED / CANCELLED	– terminal

THREE DELIBERATE DESIGN DECISIONS IN THAT TABLE

- **INCIDENT is a suspension, not a terminus.** A stuck truck that gets moving again returns to ACTIVE. Treating an incident as terminal would force operators to fake a new trip.
- **VERIFICATION_PENDING can return to ASSIGNED.** A driver who mis-photographs a truck is not stuck; the flow is recoverable.
- **DELIVERED is not the end.** CLOSED exists separately because a delivered trip still has settlement, documents and cost reconciliation outstanding.

THREE RELATED BUT DELIBERATELY DIFFERENT STATE SETS

Set	Members	Question it answers
IN_TRANSIT_STATES	ACTIVE · DELAYED	Is the truck on the road? — what Fleet Sentinel monitors, and what the partial index matches
TERMINAL_STATES	CLOSED · CANCELLED	Can anything more happen to this trip?
Assignment-holding states	wider than in-transit, narrower than non-terminal	Does this trip still tie this driver to this truck? A truck stopped at an INCIDENT, or one DELIVERED and awaiting closure, is still the truck that driver is with — but a trip still in DRAFT is ordinary dispatch work and moving the driver must not be blocked.

7.2 The eight-stage journey

1 · PLAN Shipment, cargo, endpoints geocoded. Trip and stops created in one atomic RPC. Capacity validated against <code>max_capacity_kg</code> and refused if exceeded.	2 · PREDICT Three corridors routed via OSRM. Weather sampled along each polyline. Landslide exposure evaluated. Fuel and CO ₂ computed.	3 · EXPLAIN Score, band, component breakdown, reason codes, and every factor that was <i>not</i> available, named.	4 · AUTHORIZE Eligibility gate runs before ranking. A held route needs a reviewer's single-use 30-minute token with a rationale.
5 · NAVIGATE Geometry and turn maneuvers packaged and bound to a route revision. Offline pack downloaded. Spoken guidance.	6 · TRACK GPS ingested idempotently. Freshness classified LIVE / STALE / NO_CONTACT. Progress computed along the corridor.	7 · ADAPT Conditions change → NO_ACTION, ALERT_ONLY or PROPOSE. A change applies only when a human accepts it.	8 · DELIVER Stops arrived and completed, proof of delivery captured, trip delivered then closed. Full event history retained.

7.3 Assignment verification — why it exists

When a driver is assigned a truck they have not driven before, the app requires them to photograph it and report the registration, odometer reading, fuel level and any visible damage before the trip can start. The reported values are stored alongside the record so a mismatch is a fact, not an argument.

`mismatch_flagged` routes the assignment to `MANAGER_REVIEW`. The driver is **not** blocked by an automated check — plate reading in mud, rain or darkness at 04:00 fails routinely, and a system that strands a driver over a blurry photo will be worked around within a week. The check informs a human; it does not gate the human.

8 The intelligence layer, engine by engine

READ THIS BEFORE THE REST OF THE SECTION

None of the engines below is a trained machine-learning model, and none is described as one anywhere in the code, the API or the user interface. Each is a **deterministic weighted rule with published constants**, carries a version name that says so, and has a test asserting that no `confidence`, `probability` or `model_version` field ever reaches the wire. The reason is in §9: there is no ground truth to validate a model against, and a number nobody can check is worse than a rule anybody can argue with.

8.1 route_risk — how a corridor is scored

Answers "should this truck take that road right now, and why" — not "how do I get there", which a routing provider answers.

Component	Max points	How it is computed
Rainfall	45	Scaled by intensity and by coverage across the sampled points. Thresholds: moderate at 2.5 mm/h, heavy at 7.5 mm/h. A route where rain is falling on one sample scores differently from one where it is falling on all of them.
Wind gusts	25	Notable at 40 km/h, severe at 60 km/h. Relevant for high-sided loads on ridge sections.
Duration exposure	20	Scaled against a 480-minute reference. Longer time on the road is more exposure to a condition change.
Distance exposure	10	Scaled against a 500 km reference.
Landslide	table	Points assigned by risk level from <code>domain/landslide.py</code> .

Bands: `LOW` below 30, `MODERATE` from 30, `HIGH` from 60. Duration and distance are kept in a separate `condition_points` accounting from the weather terms, so "this road is bad" and "this trip is long" are never conflated.

THREE INVARIANTS THE MODULE ENFORCES

- Stale observations do not score.** A weather reading from four hours ago describes weather that has moved on. Stale readings are counted and reported, never scored — because a stale reading that *lowers* a risk score is worse than no reading at all.
- Every unavailable factor is named in the output.** The response carries `inputs` (factor → AVAILABLE / NOT_AVAILABLE), `unavailable`, `reason_codes`, `observations_used` and `observations_stale` alongside `score`. A dispatcher who sees a bare "37/100" assumes it is complete, and that assumption is the dangerous one.
- The thresholds are this project's own.** They are chosen to be defensible and legible, they are constants in one file so they can be argued with and tuned, and they are explicitly *not* presented as an official meteorological standard.

The reason-code catalogue carries the semantic the problem statement actually needs:

```
"Hazard data for this route is unavailable.  
This is not a clear route - it is an unchecked one."
```

8.2 route_eligibility — why a refusal is not a score

Eligibility is decided *before* ranking, and separately from it. The argument is arithmetic:

```
a closed route scoring    7 + 45 = 52  
beats  
an open route scoring      70
```

A penalty can always be out-voted by the other components. A refusal cannot. So a confirmed closure produces `REJECTED` , and no score, duration saving, cargo priority or client flag may override it.

REQUIRED EVIDENCE VERSUS OPTIONAL ROADMAP FACTORS

This is the distinction that makes the policy workable. `route_risk` reports seven factors as unavailable in this build. Six of them — flood, road quality, truck restrictions, historical incidents, elevation and fuel — are **roadmap** factors: inputs nothing has ever supplied. If "any unavailable input forces review" were the rule, every route would be review-required forever on account of features that were never built, the gate would carry no information, and operators would learn to click through it.

Exactly one factor is **required safety evidence** today: **landslide**. It is the one this system exists to reason about, and the one where absence of knowledge is operationally meaningful rather than merely incomplete. So unknown landslide evidence forces `REQUIRES_REVIEW` ; unknown fuel does not.

`ELIGIBLE` means "eligible under the constraints actually checked". The module never claims it is a safety guarantee.

8.3 fuel_model — the physics

F = (B_base + K_payload x M_cargo + K_speed x max(0, v - 50)) x Distance_km
+ K_grade x max(0, delta_H)

Coefficient	Value	Meaning
B_base	0.22 L/km	Unladen flat cruising at 50 km/h — about 22 L/100 km
K_payload	0.0075 L/(t·km)	+0.75 L per 100 km per tonne of cargo
K_grade	0.00018 L/m	About 0.18 L per 1000 m of elevation gain
K_speed	0.002	Aerodynamic drag penalty above 50 km/h
CO ₂ factor	2.68 kg/L	Diesel emission factor

Calibrated for a standard multi-axle Indian freight truck, 16 t–28 t GVW. The estimate is returned with a **mode** that states how much was actually known: `CALIBRATED` · `ESTIMATED` · `BASELINE_ONLY` · `UNAVAILABLE` , and the breakdown is returned in parts — base litres, payload penalty, elevation penalty, speed penalty — so the total is checkable rather than asserted.

Only *positive* elevation change is charged. Descent does not refund fuel, because on a loaded truck it is spent on engine braking rather than recovered.

8.4 road_memory — the state machine that refuses to guess

The question this answers is not "is the road open". It is "what is the last thing anybody actually observed about this road, who observed it, and how long ago" — because the first question has no honest answer most of the time, and a system that produces one anyway will eventually route a loaded truck onto a washed-out hillside.

THE RULE THE MODULE EXISTS TO ENFORCE

Silence is not evidence. No amount of elapsed time may improve a road's status. A landslide reported on Tuesday with nothing further heard by Friday is not a repaired road; it is a landslide nobody has followed up. Every path toward `VERIFIED_OPEN` requires a positive observation. The reverse also holds: silence must not *close* a road either — inventing a closure strands cargo and teaches dispatchers to ignore the system. What silence changes is *freshness*, never *status*.

State	Meaning
VERIFIED_OPEN	Somebody observed traffic passing

State	Meaning
REPORTED_INCIDENT	Something happened; passability unknown
CLOSED	An authority said it is shut
REPAIR_REPORTED	Somebody claims it is fixed — unconfirmed
AWAITING_VERIFICATION	A repair claim has gone unconfirmed long enough that acting on it needs a decision, not a default

REPAIR_REPORTED and AWAITING_VERIFICATION carry the weight. A repair claim is exactly the kind of evidence that *looks* like an answer and is not one — it usually arrives from a source with an interest in the road being open, and it is the point where a naive system would flip to OPEN and route a truck. Keeping the claim and the confirmation as separate states is what makes the gap visible instead of assumed.

Evidence carries a **source**, and the strongest source available is not an official bulletin: it is FLEET_TRAVERSAL — one of our own trucks recorded GPS fixes along the segment and came out the other side. That is a fact about the physical world observed by this system, and it is the only evidence that can reopen a CLOSED road without an authority saying so.

8.5 monsoon_risk — exposure, with passability held separate

A weighted rule over three facts and one belief: historical recurrence (how often this stretch has actually failed), rainfall (how much has fallen recently), season (is it monsoon), and road status (what road_memory last observed).

Passability is not a score. A road an authority has declared CLOSED is not "risk 100" — it is not a road you may plan over. A closed segment scoring 100 against an open one scoring 96 would be a four-point preference for the road that exists. So passability is a separate three-valued field, and NOT_PASSABLE is a refusal rather than a high number.

UNVERIFIED is its own answer. A segment nobody has observed is not safe and not dangerous. A caller that treats UNVERIFIED as passable has made a decision this module declined to make for it.

8.6 route_recommendation — explainable comparison

Given several routes each scored by route_risk, decide which to advise and state why *in the same units the decision was made in*.

NO PERCENTAGES

A risk score is an ordinal built from this project's own weights. "33 points lower" is a statement about that scale and is checkable against the components. "54% safer" is a claim about probability of harm, which nothing here measures. Deltas are therefore reported in points, minutes and kilometres — the units the inputs arrived in — and never as a ratio.

NOTHING IS INVENTED

If a trip has one route, the answer is that route and the reason code says so. A backup corridor is not fabricated to make the comparison look richer. On a single-road corridor — which much of the North East is — one route is the truthful answer.

THE EVIDENCE-PARITY GATE

This guards a subtle failure. Suppose route A was scored *with* live weather, and route B was scored without it because the provider failed for B's sample points. B will usually score lower — purely because a factor that only ever *adds* points was missing. Recommending B would be recommending the route we know less about.

So when the available-input sets differ, **the comparison is refused** and the reason code names which fact is missing. The baseline is kept rather than a guess being dressed up as a decision.

8.7 reroute — three outcomes, not two

Outcome	Meaning
NO_ACTION	The road is not bad enough to reconsider
ALERT_ONLY	It is bad, and there is nothing better to offer
PROPOSE	It is bad, and a genuinely better road exists

`ALERT_ONLY` is the outcome that makes this honest. Most of the North East is a single corridor: when NH-715 floods there is frequently no second road, and a system that only knows how to propose alternatives has nothing to say in exactly the situation that matters most. "This road has deteriorated and there is no better option" is real information — it is what makes a dispatcher phone the driver.

NOTHING HERE MUTATES A TRIP

This module returns an assessment. It does not select a route, does not write, and is not permitted to. The route a truck is following changes only when a person accepts the proposal. A silent automatic reroute is the failure mode this design exists to prevent: a driver whose map changes underneath them on a hill road at night, with no one having decided anything.

The comparison is not re-invented here — it calls `route_recommendation.recommend`, so a reroute proposal and a planning-time recommendation cannot disagree about which road is better.

8.8 The five-stage pipeline, in order

Stage	Step	Detail
1	Corridor routing	OSRM returns geometry, duration and turn maneuvers per candidate
2	Weather sampling	Open-Meteo queried at points along the polyline; each reading carries a freshness stamp; the database connection is released first
3	Landslide exposure	Zone exposure evaluated, with <code>is_known</code> separating exposure from a current hazard report
4	Physics fuel	Litres, per-100 km rate, component breakdown and CO ₂
5	Synthesis	Weighted score, band, reason codes, component breakdown, and the named unavailable factors

If the weather provider is down the assessment still returns, with `weather: NOT_AVAILABLE` and a reason code saying so. A 503 here would mean a dispatcher sees nothing at all because one free API had a bad minute — when distance and duration are still perfectly good evidence.

9 AI and machine learning

9.1 The governance rules that apply to every model

- Every model ships with a baseline it must beat.** A model that does not beat its baseline on a held-out set is not deployed. The baseline stays in the code permanently as the fallback.
- Every model has a fallback path.** If inference fails, times out, or the model file is absent, the system returns the baseline result or an explicit `null` — never a fabricated number.
- No model output is authoritative.** Outputs inform manager decisions and are labelled as estimates.
- Provenance travels with every prediction** — `source`, `model_version`, and where applicable the baseline value alongside it.

5. **Synthetic training data is disclosed as synthetic** — in the documentation, in the UI copy and in the presentation. Metrics on synthetic data measure learnability of the generator, not real-world accuracy, and must be described that way every single time.

Rule 5 is the one most likely to be quietly broken under deadline pressure, and it is the one that would most damage the project's credibility in judging. It is written down for that reason.

9.2 Why no model is trained yet — the specific blocker

NOT "NOT BUILT YET" — NOT BUILDABLE YET FROM THE SOURCES THAT EXIST

A road-source survey was carried out. The finding that decides it: **no surveyed source publishes "this road is open again."**

- The Geological Survey of India publishes hazard forecasts and a historical inventory.
- ASDMA publishes damage and situation reports.
- IMD publishes weather and rainfall warnings.
- **Reopening is decided by PWD / NHAI / BRO and communicated ad hoc**, so nothing machine-readable carries it.

That is not merely a missing feature. It means there is **no ground truth** for "was this road passable on this date" — which is the label any landslide or route-risk model would have to be trained and validated against. A model fitted on these inputs would produce a number with no way to check it, which Rule 1 forbids (it could not be shown to beat a baseline) and which would break Rule 5 the moment it was described as anything at all.

The consequences are deliberate, and each is visible in the code:

- `road_memory.py` records evidence and enforces that silence is not evidence. That evidence log is what would one day become the labelled dataset.
- `monsoon_risk.py` is a weighted rule named `monsoon-risk-engine-v1` — not a forecast.
- `route_risk`, `route_recommendation` and `reroute` each carry a `version` naming a rule, and each has a test asserting that no `confidence`, `probability` or `model_version` field reaches the wire.

The honest sequence is: collect evidence first, label it, then ask whether a model beats the rule. Reversing that order is how the disclosure rule gets broken.

9.3 The four models specified in full but not trained

FUEL AI

Aspect	Specification
Problem	Regression: route + truck + load + conditions → litres consumed, so routes can be compared on true cost. Distance-only estimation is specifically wrong in NER — gradient is the dominant term this model exists to capture.
Features	<i>Route</i> : distance, cumulative ascent and descent, mean absolute gradient, % distance above 4% grade, max gradient, road-class mix. <i>Truck</i> : type, axle count, kerb weight, engine capacity, year, baseline mileage. <i>Load</i> : payload kg, load ratio. <i>Conditions</i> : monsoon flag, forecast rainfall, expected mean speed, road quality index, night fraction. <i>Historical</i> : truck kmpl mean and std, driver kmpl mean.
Deliberately excluded	Driver identity as a raw categorical. It would let the model encode individual drivers, and a fuel estimate that varies by <i>who</i> is driving invites disciplinary misuse of an advisory number. Aggregate driver history is included; identity is not.
Baseline	The physics heuristic already shipping. If the ML model cannot beat it, the ML model has no reason to exist, and the honest outcome is to ship the baseline and say so.

Aspect	Specification
Candidates	LightGBM (primary), XGBoost (comparison), Ridge on engineered physics features (interpretable third point). Gradient boosting rather than a neural network: small tabular dataset, and a tree model can be explained to a judge in one sentence.
Training data	No real NER truck fuel logs exist. A physics-informed synthetic generator is specified — real NER road geometry, real SRTM elevation, tractive-effort model, then realistic noise and per-truck bias. It can demonstrate the pipeline, the serving path, the fallback behaviour and that the model recovers a known non-linear relationship better than a linear baseline. It cannot produce a real-world accuracy claim.
Evaluation	MAE, RMSE, MAPE, and MAE relative to baseline MAE (the headline). Segmented by gradient band and load ratio — a model that only wins on flat terrain is uninteresting. Split by route corridor, never randomly by row, to avoid leaking near-identical segments.
Failure handling	Model missing → baseline formula. Inference over 500 ms → baseline, log timeout. Feature unavailable → baseline, never impute silently. Prediction below 0 or above 3× baseline → reject, use baseline, log for review. Baseline mileage also null → return <code>null</code> , UI shows "unavailable".

ROUTE RISK AI

Aspect	Specification
Problem	Score a candidate route's likelihood of disruption, to <i>rank</i> routes that have already passed the deterministic passability filter. Ranking, not gating.
Notable feature	<code>hours_since_last_confirmed_passage</code> , derived from our own GPS data — the feature most likely to carry real signal, because a road another truck traversed two hours ago is empirically open.
Baseline	Rule table: IMPASSABLE incident within 5 km → 1.0; heavy-rain warning in a known landslide zone → 0.7; monsoon season → 0.4; else 0.1.
Evaluation	PR-AUC, not ROC-AUC — disruptions are rare and the negative class dominates. Brier score for calibration. Recall at a fixed low false-positive rate, because under-predicting a landslide is far worse than over-predicting one.
Hard limit	A risk score never blocks a route on its own. Only a confirmed IMPASSABLE incident does that, deterministically.

ETA CORRECTION

Predicts the **correction** to the routing engine's ETA rather than the ETA itself: `predicted_eta = engine_eta + model_correction`. This is deliberate — the correction target is small and well-behaved, the model inherits the engine's road knowledge instead of relearning it, and if the model fails, `correction = 0` degrades exactly to the engine's own estimate. Corrections beyond ±50% are clamped and logged. Evaluated on MAE in minutes and on the *percentage of trips within ±15% of actual*, which is closer to what a manager cares about than raw MAE.

VEHICLE PHOTO VERIFICATION

PaddleOCR text detection on the plate region, normalised (uppercase, strip separators, disambiguate O/0 and I/1), fuzzy-matched against the expected registration. The evaluation set must include deliberately poor images, because that is what a 04:00 depot photo looks like.

The important part is the failure handling. `unreadable` is the expected common case, not an error — mud, rain and darkness are normal. OCR never blocks a driver and never auto-rejects an assignment; a mismatch sets `mismatch_flagged` and routes to manager review while the driver proceeds. If PaddleOCR is unavailable, verification falls back to manager photo review and the workflow is unchanged. It is a convenience layer over a manual process that must work without it.

9.4 The metrics table — empty by design

Model	Version	Trained	Baseline	Model	Improvement	Data
Fuel	—	not trained	—	—	—	—
Route Risk	—	not trained	—	—	—	—
ETA	—	not trained	—	—	—	—
Photo verification	—	not evaluated	—	—	—	—

This table is filled in only from a real evaluation run, with the date and the data type stated. Until then it stays empty, and nothing anywhere in the project — UI, README or presentation — claims a number that is not in it.

9.5 The language model — what it is allowed to do

This is not one of the models above and it never enters their decision paths. Sections 8 and 9.3 describe deterministic or predictive systems with safety authority; this is a language model that **rewords facts those systems have already produced**.

Surface	Given	May produce
Assistant	Facts computed from the signed-in driver's own trip	A restatement of those facts
Safety	The bundled, reviewed guidance the driver is already reading	A rewording of that text
Translate	One short message plus a language pair	A translation, labelled as machine output

WHAT IT MAY NEVER DO

- Decide route eligibility, a hazard refusal, capacity, a review authorisation or any lifecycle transition. It has no write path to any of them.
- Turn an unavailable fact into a value. `unavailable` travels through the prompt as the word, and the prompts forbid guessing a distance, a time, a road name, a phone number, or whether a road is open.
- Add medical advice, medicine names or doses, a road-is-clear verdict, or a promise that help is coming. These are pinned by tests.
- Replace the offline guide, the deterministic assistant or the reviewed phrase pack. All three still work with no model, no server and no internet, and each screen states which of the two the reader is looking at.

BOUNDARIES THAT ARE STRUCTURAL, NOT CONVENTIONS

Boundary	Mechanism
Driver-scoped	Every route takes <code>CurrentDriver</code> and no id, so there is no parameter that could select whose trip is described
Local-only (local path)	<code>looks_local()</code> refuses a non-loopback host unless <code>AI_ALLOW_NON_LOCAL_HOST</code> is deliberately set — otherwise pointing the URL at a hosted endpoint would silently send a driver's trip off the machine
Bounded	One concurrent generation process-wide, a prompt-length cap and an output-token cap, so a local decode cannot starve GPS ingestion
Stateless	No conversation is stored; each question carries its own facts
Key isolation	Provider keys live server-side in the edge function. No client holds one.
Injection defence	Adversarial prompt guards and chain-of-thought stripping before any text reaches the driver

PROVIDER STATUS MODEL

One `buildStatus()` function serves both status routes — previously each assembled its own object and they disagreed, because two implementations of one question always eventually differ. States are computed from what the providers actually *did*, not from key presence:

State	Meaning
OFFLINE	No provider key configured at all
CONFIGURED	A key exists; nothing attempted yet in this isolate
USABLE	The last attempt produced a real generated answer
QUOTA_EXHAUSTED	The last attempt was refused for quota
FALLBACK	The primary failed and the backup engine answered

`available` is true only for `USABLE` and `FALLBACK` — because a generated answer is obtainable in the fallback case, just not from the primary.

9.6 Fleet Sentinel is not machine learning — and never may be

This is stated explicitly and permanently. Fleet Sentinel is deterministic application logic: a scheduled query, threshold comparisons and a state machine. Three independent reasons no model may enter its decision path:

- 1. **Explainability under scrutiny.** After a real incident, "the truck had not moved more than 50 m in 60 minutes and the driver did not answer in 30" is defensible. "The model scored 0.83" is not.
- 2. **No training data, and asymmetric error cost.** A missed genuine emergency is catastrophic; a false positive costs one driver one button press.
- 3. **Availability.** The safety path must not have a dependency that can be down.

Where ML may eventually assist — strictly outside the decision path — is reducing false positives by learning genuine rest stops from historical patterns. Even then it would only *suggest geofences for manager approval*; it would never suppress a check directly. A language model may summarise a resolved incident for a report **after** resolution; it may not participate in detection, escalation or resolution.

10 GPS, telemetry and truthfulness

10.1 Freshness classification

Label	Condition	What the UI shows
LIVE	fix age ≤ 90 s	Solid marker, live speed
STALE	90 s < age ≤ 600 s	Faded marker, last-known position with its age
NO_CONTACT	age > 600 s	Distinct no-contact state — the Sentinel's input
NO_LOCATION	no fix at all	Explicitly "no location", never a default coordinate

10.2 Adaptive tracking on the device

Parameter	Value	Reason
TRACKING_MOVING_INTERVAL_SECONDS	10	Resolution while the truck is moving
TRACKING_STATIONARY_INTERVAL_SECONDS	60	Battery preservation while parked
TRACKING_STATIONARY_DISTANCE_M	30	Movement threshold that switches between the two

Parameter	Value	Reason
TRACKING_BATCH_SIZE	6	Normal upload batch — few requests, low latency
TRACKING_MAX_BATCH	500	Ceiling when draining a queue after a dead zone
TRACKING_QUEUE_LIMIT	500	Bounded offline buffer — a long outage cannot exhaust device storage

10.3 Anomaly and rejection handling

Condition	Threshold	Treatment
MOCK_LOCATION	device flag	Flagged and stored , not discarded — a spoofed fix should be visible in the record
IMPLAUSIBLE_SPEED	200 km/h	Flagged as an anomaly
POOR_ACCURACY	500 m	Flagged; the fix is kept with its accuracy so downstream logic can weigh it
STALE	—	Rejected at ingest with a named reason
FUTURE_TIMESTAMP	—	Rejected at ingest with a named reason

THE TRUTHFUL-TELEMETRY CONTRACT

Telemetry is either live from the device sensors or honestly reported as unavailable. There is no placeholder coordinate anywhere in the system, no interpolated position presented as observed, and no default speed. A test suite explicitly checks position truthfulness and honest clock aging — including that a displayed age keeps increasing rather than freezing at the last render.

11 Offline and resilience design

Connectivity loss is the normal operating condition on these corridors, not an exception to be handled defensively. The design assumption is inverted from a typical fleet product.

Mechanism	Behaviour
Offline package	<code>GET /api/driver/me/trip/offline-package</code> bundles everything needed to keep working with no network. If it cannot be built, it refuses rather than returning an empty package that would look like "there is nothing to show".
Navigation package	<code>GET /api/driver/me/trip/navigation</code> returns geometry plus turn maneuvers, bound to a <code>route_revision</code> .
Route-revision binding	The client discards a package whose <code>route_id</code> no longer matches, so a reroute landing mid-request can never draw a new line with old turns.
Cached fallback	If a live fetch fails, the stored package is used and labelled <code>source: CACHED</code> with the capture time — never presented as live.
Reroute-race safety	A <code>matches()</code> check discards geometry whose route id changed mid-flight.
GPS buffering	Fixes queue on the device up to the queue limit and replay on reconnection. Idempotency on <code>(trip_id, device_fix_id)</code> makes replay safe.
Offline emergency guide	Bundled with the app. Dial shortcuts for 112, 108 and 1033 work with no data connection.
Offline phrasebook	A reviewed phrase pack in Assamese, Bengali, Hindi and English — works with no model, no server and no internet.

Mechanism	Behaviour
Rest tracking	A 45-minute mandatory break tracker runs entirely on the device.
Reason-code i18n	The backend sends codes, not sentences; the client renders the sentence from <code>i18n/reasonCodes.json</code> . This is what delivers Hindi and Assamese explanations without a translation model in the loop.

HONEST DEGRADATION, WORKED EXAMPLE

Address search depends on a geocoding provider. When that provider fails, the manager UI does not show a spinner or a silent empty list — it renders `PROVIDER_ERROR` and offers a working **"Choose on map"** alternative that yields the same coordinate. The degraded path is named, and the workflow still completes.

12 Safety: Fleet Sentinel and escalation

12.1 What it does

A scheduled query over trips in `IN_TRANSIT_STATES` (`ACTIVE` and `DELAYED`, matched by the partial index `ix_trips_active`), using `ST_DWithin` to detect a truck that has not moved outside an approved stop, combined with timer comparisons on driver response.

Aspect	Position
Trigger	A stationary active truck outside an approved stop, plus driver silence
Method	Scheduled query + <code>ST_DWithin</code> + timer comparison. No inference anywhere.
Input	<code>gps_points</code> , <code>trips.status</code> , geofences, configured thresholds
Output	State transitions and a populated SOS briefing to the manager
Driver interaction	One question, one very large button: Are you safe?
False-alarm handling	Two-step confirmation and a cancel path, because a false positive costs one button press while a missed emergency does not
Failure of the scheduler	Alarmed loudly. A missed safety check is treated as a Sev-1. There is no silent degradation here.
Fallback	None needed — there is nothing to fall back from. It is the rule.

12.2 What the manager receives

Not an alert that says "check on truck 12". A briefing containing the last known position and its age, the selected corridor, the stop sequence and which stops are complete, the cargo, the driver's name and phone, their emergency contact, the truck registration, and the trip's recent event history. Everything needed to act, assembled before the manager asks for it.

13 Security, permissions and privacy

13.1 Authentication

- **Argon2id** (RFC 9106) password hashing.
- **15-minute access tokens, 30-day refresh tokens.**
- **Rotating refresh with reuse detection.** Every refresh issues a new token and revokes the old one. Presenting an already-replaced token revokes the entire family — the standard response to a stolen refresh token.
- Only the token *hash* is stored, with the user agent and IP that presented it.

- **Rate limiting:** login 20/min per IP and 10/min per identifier; refresh 60/min per IP; 60-second windows. The per-identifier limit is what stops a distributed attack on one known account.

13.2 Authorisation

Four roles against approximately 28 named permissions. Permissions are constants, not strings scattered through handlers.

Group	Permissions
Drivers	<code>driver:read</code> · <code>driver:create</code> · <code>driver:update</code> · <code>driver:deactivate</code> · <code>driver:read_sensitive</code>
Trucks	<code>truck:read</code> · <code>truck:create</code> · <code>truck:update</code> · <code>truck:retire</code>
Assignments	<code>assignment:read</code> · <code>assignment:create</code> · <code>assignment:end</code> · <code>assignment:review</code> · <code>assignment:verify_own</code>
Shipments / trips	<code>shipment:read</code> · <code>shipment:create</code> · <code>trip:read</code> · <code>trip:create</code> · <code>trip:dispatch</code> · <code>trip:cancel</code> · <code>trip:close</code>
Routes	<code>route:read</code> · <code>route:plan</code> · <code>route:select</code> · <code>route:review_authorize</code>
Driver-scoped	<code>trip:execute_own</code> · <code>location:submit_own</code>
Fleet / audit	<code>fleet:location_read</code> · <code>audit:read</code>

The `_own` suffix is load-bearing. A driver has `trip:execute_own` and `location:submit_own`, never the unscoped versions — so there is no request shape in which a driver can act on another driver's trip. `route:review_authorize` is separated from `route:select` precisely so that the person who releases a held route need not be the person who dispatches it.

13.3 Data protection

Control	Detail
Row Level Security	Enabled on every table with no policies. Supabase publishes <code>public</code> through its Data API, so a table without RLS is readable by anyone holding the anon key.
No client credentials	The database password lives only in <code>backend/.env</code> inside <code>DATABASE_URL</code> . It is never logged and never returned by an endpoint — <code>/ready</code> exposes only the provider name.
Service-role key	<code>SUPABASE_SERVICE_ROLE_KEY</code> bypasses all row-level security. It is not needed, is absent from <code>.env.example</code> , and must never appear in a client, a bundle, a log or git.
Public-by-prefix	<code>VITE_*</code> and <code>EXO_PUBLIC_*</code> are inlined into client bundles and are therefore public by construction. No secret may use those prefixes.
Startup refusal	The backend refuses to start with the placeholder <code>SECRET_KEY</code> when <code>APP_ENV</code> is not development.
Provider refusal	<code>DATABASE_PROVIDER=supabase</code> with a localhost URL is rejected at startup. There is no automatic fallback — if the configured primary is unreachable, <code>/ready</code> returns 503 rather than quietly serving a different database.
Transport	<code>DB_REQUIRE_SSL</code> defaults to true.
Audit	Append-only <code>audit_logs</code> with actor, action, entity, before / after JSON, reason and IP. The actor foreign key is restrictive so history cannot be orphaned by deleting a user.
Test-database guard	A <code>do_connect</code> listener on SQLAlchemy's <code>Engine</code> class with an exact-match allowlist of one target. A wrong database is <i>never dialled</i> , rather than merely discouraged, and collection additionally refuses the run with one clear message. Refusal messages are tested not to leak credentials.

Control	Detail
Release gate	<code>check-release-config.mjs</code> fails the EAS build if a LAN backend URL is configured, so a laptop-dependent APK cannot ship. This is the only control that actually prevents it — a convention would not.

14 The two client applications

14.1 Manager web console

Page	What the manager does there
LoginPage	Email and password; role-aware landing
FleetPage	The command picture: live MapLibre map, 4-column KPI bar (Active / Moving / Stale / No-contact), planned corridor drawn distinctly from the observed GPS breadcrumb, slide-out truck drawer with cargo utilisation, live speed and a direct call action
TripsPage	Plan a shipment and trip atomically, dispatch, cancel, close; refuses an unlocated endpoint
DriversPage	Roster, documents and expiry states, deactivation
TrucksPage	Fleet register, capacity, documents, maintenance, retirement
AssignmentsPage	Driver ↔ truck pairing, verification review, ending assignments
ReviewPage	The authorised-reviewer queue for held routes
SystemPage	Backend health and provider status

The centrepiece is `RouteRiskComparison` — a three-corridor stepper showing each candidate's distance, ETA, risk band and score, the component breakdown, the reason codes, the **named evidence gaps**, and a fuel comparison bar chart from the physics model. `TripRouteReview` is the authorisation modal that issues the single-use token.

14.2 Driver Android app

Screen	What the driver does there
LoginScreen	Sign in; token stored in the hardware-backed keystore
AssignmentScreen	Truck verification — photo, registration, odometer, fuel level, damage notes
MapScreen	The cockpit: dark industrial theme, high-contrast next-turn banner at the top, floating circular controls on the right (alt route, mute, fit route, recentre), live speedometer bottom-left, collapsible drawer with stops, cargo weight and POI discovery
TripScreen	Stop list, arrive and complete actions, trip progress
SafetyScreen	Offline emergency guide, 112 / 108 / 1033 dial shortcuts, 45-minute rest tracker
AssistantScreen	Four sub-modes of trip-scoped question answering
TranslateBox	Single-message translation, labelled as machine output, with audio
PhrasebookScreen	Reviewed offline phrase pack

Roadside services are discovered spatially across three modes — Near Me, Along Route and Viewport — covering emergency services, tyre repair, hotels, fuel and rest stops.

14.3 Interaction certification

Every button, tab and toggle on both clients is audited to be in exactly one of three states: **WORKING**, **DISABLED_WITH_REASON** (visibly disabled with the reason stated), or **REMOVED**. There are no inert controls that look actionable. This is recorded as a certified audit rather than an aspiration, because a dead button in a demo is indistinguishable from a broken product.

15 Verification and testing

Suite	Result	Environment
Backend	959 passed, 5 skipped, 0 failed (140 s)	Isolated PostgreSQL 17 + PostGIS cluster on 127.0.0.1:55432/ner_logistics_test , target-guarded
Manager web	124 passed (12 files)	Vitest + Testing Library + jsdom
Driver app	490 passed (36 files)	Vitest over react-native-web + jsdom
Type checking	Clean on both clients	tsc
Production build	Succeeds	Vite

The 5 backend skips are named and legitimate: one non-Windows event-loop test, and four destructive migration tests gated behind RUN_DESTRUCTIVE_MIGRATION_TESTS=1 because they downgrade to base and drop every table.

15.1 What specific suites pin down

Suite	Guarantee
test_db_target_guard.py	The run fails closed against any remote or unverified database, and a refusal does not leak credentials
test_golden_path_e2e.py	One complete manager + driver operational cycle, end to end
test_route_recommendation.py	Deterministic comparisons, evidence integrity, and that no confidence-like field reaches the wire
test_route_risk.py	Scoring, banding, stale-observation handling and unavailable-factor reporting
test_fuel_model.py	Unladen baseline, laden payload, elevation grade penalty
test_trip_execution.py	Lifecycle transitions and stop handling
test_concurrency.py	Behaviour under simultaneous writes
test_schema_drift.py	The ORM and the migration head agree
test_gemini_ai.py	Prompt-injection defence, medical-diagnosis refusal, rate limiter
MapScreen.test.tsx	Position truthfulness, honest clock aging, permission revocation
TripsPage.test.tsx	Atomic shipment + trip creation; unlocated endpoint refusal
FleetMap.worker.build.test.ts	The production build actually emits the MapLibre worker
geminiAiHandler.test.ts	Reasoning-token stripping and per-mode prompt specialisation
queueStorage.test.ts	Offline GPS queue persistence and bounded growth

15.2 Field verification

An installable Android APK — VersionCode 9, build 3730b06b , 71.29 MB, SHA-256 228a1d3f5a7e3d2705fd818ebf6d7e275f84d74b61775133fbddf1d7ced437f9 — was produced through a hermetic EAS

build and exercised end to end against the hosted PostGIS database in `ap-south-1` . It passed a static security audit.

16 Deployment and environments

Component	Where it runs	State
Database	Supabase, ap-south-1	Live — PostgreSQL 17.6 + PostGIS 3.3
Edge functions	Supabase Deno runtime	Deployed — <code>gemini-ai</code> verified HTTP 200
Manager web	Static host	Built; descriptors exist for both Netlify and Vercel
Driver app	Android APK via EAS	Released — VersionCode 9
FastAPI backend	Container	READY, NOT DEPLOYED — <code>Dockerfile</code> and <code>render.yaml</code> written; no hosting account was available on the build machine
Local database	WSL2 Ubuntu, PostgreSQL 18 + PostGIS 3.6	Optional offline mode, opt-in only

16.1 Database provider selection

```
DATABASE_PROVIDER=supabase -> uses DATABASE_URL      (rejected if it names a local host)
DATABASE_PROVIDER=local    -> uses LOCAL_DATABASE_URL (requires scripts/db-start.ps1)
```

There is **no automatic fallback** between them. If the configured primary is unreachable, `/ready` returns 503 rather than quietly serving a different database — because silently serving the wrong data is worse than being visibly down. Alembic follows the same setting, so it can never migrate one database while the application serves another.

16.2 Two operational notes worth recording

- **Use the session pooler on port 5432**, not the transaction pooler on 6543. The direct host is IPv6-only while the pooler is IPv4; and the transaction pooler breaks the prepared statements `psycopg` uses and is unsafe for Alembic DDL.
- **Start the backend with `run.py` , not `uvicorn` directly.** On Windows the `asyncio` selector policy must be set before `uvicorn` creates its event loop, or every database call fails with a `psycopg InterfaceError` .

17 Known gaps, stated honestly

This section exists because a project that claims no gaps is not being audited. Each item below was found by our own review, not by an external one.

Gap	Status	Detail
Five risk factors unavailable	BY DESIGN	Flood, road quality, truck restrictions, historical incidents and terrain are declared <code>NOT_AVAILABLE</code> rather than computed. Multi-factor accessibility intelligence today rests on weather, landslide exposure and route exposure — and the demo narration must say so.
No hosted backend	READY	Container config is written. Until it is deployed, manager write operations not covered by a Supabase RPC require the local backend.
No trained model	BLOCKED BY DATA	See §9.2. Deliberate, documented, and the blocker is a data-availability fact, not an engineering shortfall.

Gap	Status	Detail
Evidence-mode taxonomy incomplete	PARTIAL	LIVE , UNAVAILABLE and UNVERIFIED exist. SOURCE_BACKED_SNAPSHOT , SIMULATED and FIXTURE do not — so a simulated demo hazard event currently has no vocabulary to label itself with.
Two hosting descriptors	FIX BEFORE PUBLISH	Netlify and Vercel configs both exist alongside render.yaml . One manager descriptor should be deleted so the app cannot be deployed twice with different environment variables.
.runtime/ ignore rule	FIX BEFORE PUBLISH	Protected by .git/info/exclude , which is local-only and never committed. Anyone cloning the repository and recreating .runtime/ gets no protection. Move it into the tracked .gitignore .
GPS-loss and provider-failure runtime testing	CODE DONE	Handled in code and unit-tested; not yet exercised as a runtime scenario on a device in a real dead zone.

ONE RESOLVED ISSUE WORTH RECORDING, BECAUSE IT SHOWS THE STANDARD

An earlier build of the shipped manager path derived a route's risk score from the route's *kind* rather than from evidence, and reported invented supporting figures alongside a claim that the weather and landslide datasets had been consulted. This was caught in our own audit and **deleted, not demoted to a fallback**. Route risk now comes from the real path, and when that path is unreachable every candidate returns `score: null` , `band: UNASSESSED` , `eligibility: NOT_ASSESSED` and the reason code `ACCESSIBILITY_ASSESSMENT_UNAVAILABLE` . All the fabricated strings were confirmed absent from the rebuilt production bundle. A fabricated number is worse than a missing one, because a missing one is visibly absent.

18 Roadmap

Horizon	Work
Immediate	Deploy the FastAPI container so every manager write is hosted; delete the duplicate hosting descriptor; move .runtime/ into the tracked ignore file; complete the evidence-mode taxonomy so simulated demo events can label themselves.
Near term	Convert route selection and reroute acceptance to server-side RPCs, matching plan_trip and dispatch_trip . Runtime-verify GPS loss and provider failure on a device. Field-test the offline package in a real dead zone.
Data layer	Ingest GSI landslide inventory, ASDMA situation reports and IMD district warnings on a schedule; build the closure/reopening evidence pipeline; begin accumulating FLEET_TRAVERSAL observations as the first genuinely labelled NER passability dataset.
Model layer	Once labels exist: train the fuel model against the physics baseline; train ETA correction against the static multiplier; train route risk against the rule table. Deploy only what beats its baseline on held-out data, and publish the metrics table with the evaluation date and data type.
Scale	Pilot the Guwahati – Shillong – Silchar corridor pair, then per-state rollout on the same schema. Nothing in the architecture is NER-specific except the hazard feeds, so the same system serves Himalayan, Western Ghat and island logistics.

19 Glossary

Term	Meaning in this system
Accessibility intelligence	Modelling the road network as changeably passable rather than fixed, and revising which corridors are usable from live evidence
Corridor	A candidate route as an actual road geometry, not a straight line between endpoints
Eligibility	Whether a route may be taken at all — decided before, and separately from, how it scores
Evidence parity	The requirement that two routes were scored from the same available inputs before they may be compared
Fleet Sentinel	The deterministic safety net that escalates on driver silence
Freshness	How old an observation is — tracked separately from what it says, and never allowed to change what it says
Road memory	The five-state record of what was last observed about a stretch of road, by whom, and when
Reason code	A machine-readable explanation token; the client renders the sentence, so explanations are multilingual without translation
Route revision	The identifier binding a navigation package to one specific corridor, so a mid-flight reroute cannot draw a new line with old turns
Silence is not evidence	The rule that no elapsed time may improve a road's status; only a positive observation can
Single-use authorisation	A 30-minute reviewer token bound to an evidence digest, consumed on selection, attributed and revocable
UNASSESSED	The honest state of a route whose risk could not be evaluated — distinct from, and never rendered as, low risk

DOCUMENT PROVENANCE

Compiled 9 September 2026 from the working tree of `ner-ai-logistics`: `backend/app/domain/**`, `backend/app/models/**`, `backend/app/api/**`, `backend/alembic/versions/**`, `manager-web/src/**`, `driver-app/src/**`, `supabase/functions/**`, and the audit reports in `docs/` — principally `PRODUCT_VISION.md`, `AI_MODELS.md`, `SIH26002_GAP_MATRIX.md`, `SIH26002_FINAL_PROGRESS.md` and `CLAUDE_RELEASE_PREFLIGHT.md`. Test counts are from the verified runs of 8 September 2026.